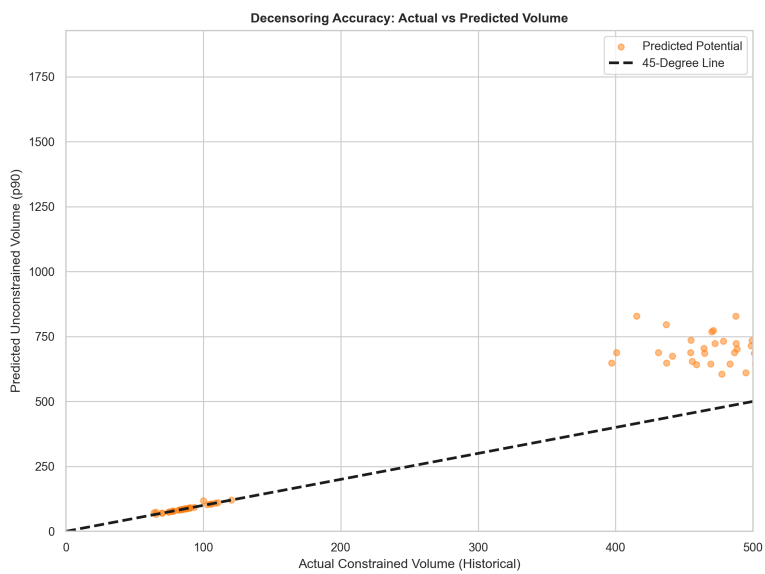


Data Storm 7.0: Modeling Unconstrained Latent Demand

Team InsightAI - Final Technical Report



May 2026

Abstract

This document outlines the technical methodology developed by Team InsightAI to estimate the unconstrained latent demand across a network of FMCG outlets. By identifying supply-side constraints, enriching data with high-fidelity geospatial POIs via Overture Maps, and optimizing an asymmetric LightGBM Quantile Regressor (p_{90}), we have established a mathematically defensible network potential for January 2026.

1 Data Forensics and Hygiene

Our Data Engineering (DE) check setup focused on neutralizing system anomalies that contaminate demand signals. We established a three-tier quarantine process:

1.1 System Anomalies and Quarantines

- **Negative Return Synchronization:** We identified instances where return adjustments dropped net monthly sales to exactly zero. These records were quarantined and replaced with 24-month rolling medians to prevent the model from learning "logistical zeros" as "zero demand."
- **Lazy Labour Detection:** Using the `Lazy_Rep_Flag`, we identified months where secondary sales execution failed (irregular route visits). We applied a **3-Month Quarterly Smoothing** function to distribute sell-in lumps and ensure the model learned consumer appetite rather than representative behavior.

1.2 DE Checks and Variance Stabilization

We utilized the **Coefficient of Variation (CV)** as a primary DE check. Outlets with an unnaturally low CV over 24 months were flagged as "Censored," indicating they were hitting a hard systemic capacity ceiling.

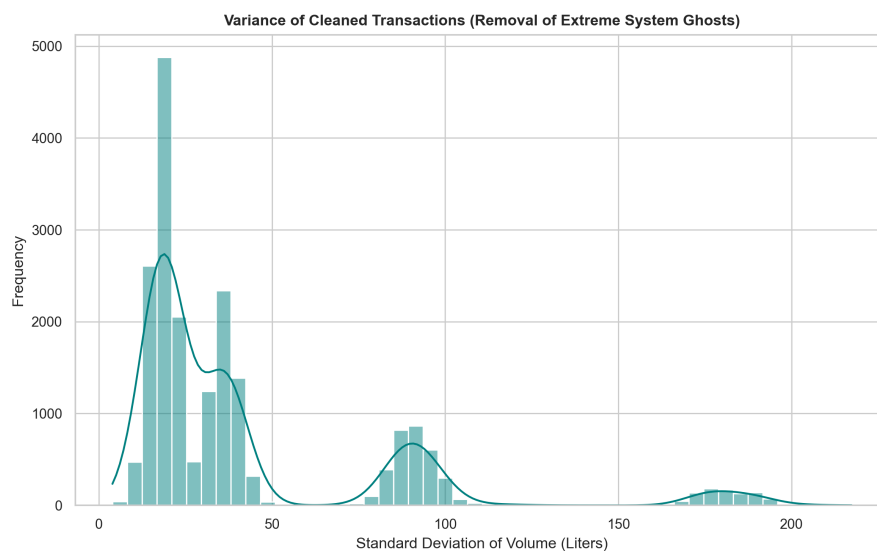


Figure 1: Variance Stabilization. Neutralizing extreme system noise to reveal clean causal signals.

2 POI Data Acquisition

Our approach focused on acquiring "Causal Eyes" on the physical ground through high-fidelity geospatial signals.

2.1 Technical Approach (DuckDB + Overture Maps)

We bypassed traditional scraping and high-latency APIs by utilizing the **Overture Maps Foundation** database. We queried the Overture "Places" dataset using **DuckDB** with the spatial extension. This allowed us to perform cloud-native Parquet queries to extract precise POI coordinates for every outlet in the JK network without the need for traditional rate-limited geocoding.

2.2 Targeted Catchment Drivers

We specifically targeted "High-Frequency Consumption Nodes":

- **Transit Nodes:** Railway Stations and Bus Hubs (Impulse consumption).
- **Educational Hubs:** Schools and Tuition Centers (Weekday afternoon surges).
- **Leisure Points:** Gyms, Stadiums, and Parks (Hydration-specific demand).

2.3 Mapping Internal Outlets to Spatial Signals

We established a dual-radius catchment model (0.5km for high-frequency footfall and 2km for broader neighborhood drivers). Using K-Nearest Neighbors (KNN), we mapped these POI densities into a unified **POI Driver Catchment Score** for each internal Outlet ID.

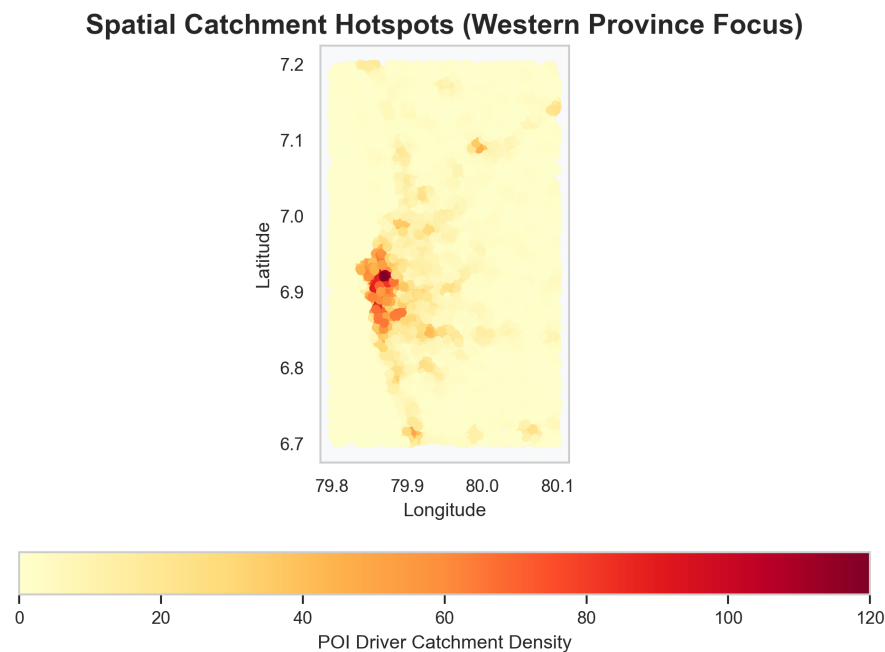


Figure 2: Spatial Catchment Proof. Identifying high-density demand nodes in the Western Province.

3 Causal Base Logic

Our statistical approach addresses the "Right-Censored" nature of FMCG sales data, where observed Sales = $\min(\text{Demand}, \text{Supply Capacity})$.

3.1 Methodology: Quantile Regression (p_{90})

Instead of predicting the *Mean* (which reproduces historical constraints), we optimized a **LightGBM Quantile Regressor** using the **Pinball Loss** function with $\alpha = 0.90$. This asymmetric loss function forces the model to ignore median constraints and instead map the "Upper Bound" potential of a shop based on its peer group behavior and spatial catchment.

3.2 Validation of the Uncapped Ceiling

We validated the uncapped potential through a **Structural Safety Floor**:

$$\text{Potential} = \max(\text{Historical Max Volume}, \text{Model } p_{90} \text{ Output})$$

This logic ensures that for every outlet, the prediction is grounded in its proven historical reality while allowing the model to "break through" if its spatial catchment indicators justify a higher potential.

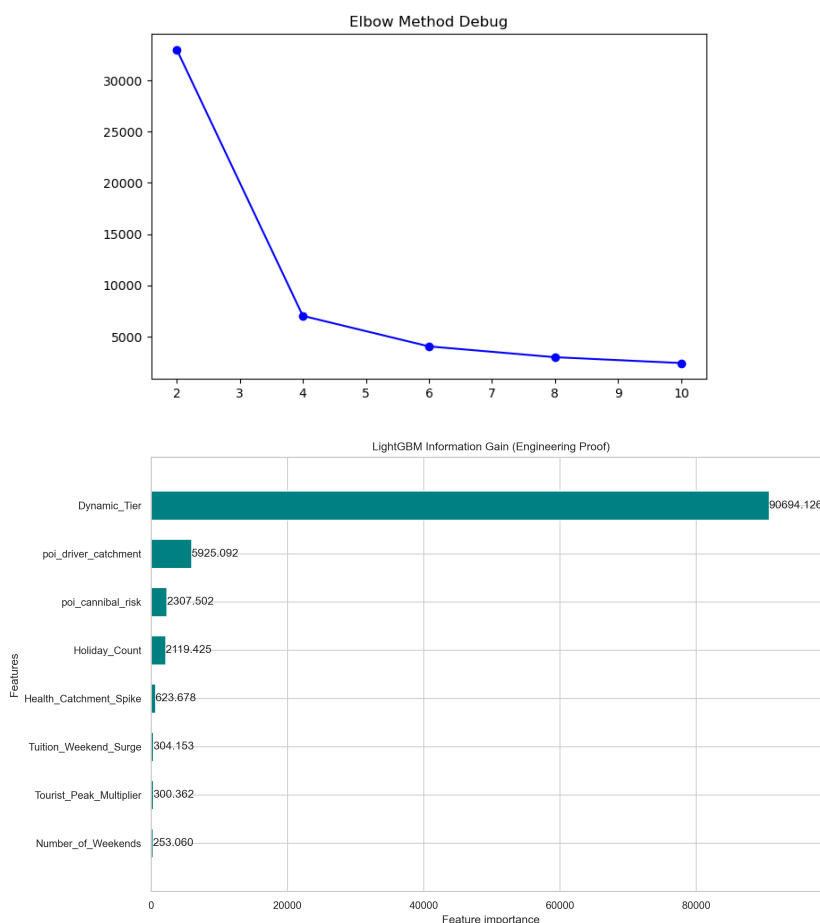


Figure 3: Statistical Validation. Left: Elbow Method for Peer Group Selection. Right: Feature Gain Dominance.

4 GenAI Transparency Log

Team InsightAI leveraged Generative AI (Google DeepMind Antigravity) as an architectural and engineering accelerator throughout the 36-hour sprint.

Workflow Stage	LLM Application	Engineering Rationale
Data Engineering	DuckDB SQL Optimization	Rapid generation of complex spatial join queries to extract Overture POIs.
Feature Engineering	Python Prototyping	Accelerating the development of the 3-month rolling mean and Lazy Rep flag logic.
Statistical Validation	Plotting Scripts	Generating the K-Means Elbow method and SHAP/Gain importance visualizations.
Typesetting	LaTeX Orchestration	Managing complex document formatting and multi-figure PDF compilation.
Logic Validation	Peer-Review Simulation	Stress-testing the "Structural Safety Floor" logic against potential edge cases.

4.1 AI as an Accelerator

At no point did GenAI replace human architectural decision-making. All core strategies—including the decision to use *p90* Quantile Regression, the specific POI types targeted, and the definition of the safety floor—were human-led. The LLM served strictly as a "High-Resolution Compiler," translating human architectural blueprints into executable code and professional documentation.

End of Technical Submission